

2026 Secondary 2 Notes

Unit 07: Rate and Speed

Name _____ () Class _____

Unit EUs

Students will be able to understand that:

1. real-world motion can be represented using distance-time and speed-time graphs.
2. the gradient and area under these graphs provide key insights into motion, such as speed, acceleration, and distance travelled.
3. the relationships between distance, speed, and acceleration allow us to analyze and predict motion effectively in various real-world contexts, such as driving, running, and falling objects.

Unit EQs

1. How can graphs help us describe and analyze motion?
2. What do the gradient and area under a graph tell us about speed, acceleration, and distance?
3. How are distance, speed, and acceleration related, and why does this matter in real life?

Unit Objectives

1. Calculate the gradients and the area under linear graphs.
2. Recognise that the gradient of a distance-time graph represents the average speed.
3. Recognise that the gradient of a speed-time graph represents the acceleration.
4. Recognise that the area under a speed-time graph represents the distance travelled.
5. Explain the relationship between distance, speed and acceleration with respect to time.
6. Explain the relationship between water level with respect to time.

Reference

Think! Mathematics New Syllabus Mathematics 3A, 8th Edition Chapter 5

Teaching To The Big Idea

Lesson sequence in the unit								
Student Learning Outcomes	Dimensions (Please tick the appropriate boxes)							
	FUNCTIONS F	INVARIANCE I	NOTATIONS N	DIAGRAMS D	MEASURES M	EQUIVALENCE E	PROPORTIONALITY P	MODELS M
Rate and Speed				✓	✓			

ACTIVITY 1

(Authentic Learning - using actual typing experience to develop a sense of rate!)

Go to <https://thetypingcat.com/typing-speed-test/1m>

Try out their typing speed test.

My typing speed is 29 WPM.

What does WPM stand for? words per minute.

Why is WPM a rate? number of words and time taken. ←



Rate is a measure combining **MEASURES** of two attributes and the unit used is derived from the units of the two **MEASURES**.

Checklist for Self-Assessment (Please tick ✓ at the appropriate boxes)

Levels	Characteristics	Knowledge and Skills	I need help. I need to see an example.	I can do this on my own.	I can do this on my own as well as explain my solution to my classmate or teacher.
Level 0	Reproduction of facts, rules, formulae; No explanations required	Converting units for speed			
Level 1	Procedural tasks without connections, Algorithmic in nature; No explanations required	Deriving gradient on a kinematic graph			
Level 2	Procedural tasks with connections, Algorithmic in Nature; explanation	Deriving speed time graph with reasoning			
Level 3	Procedural tasks with connections, Algorithmic in Nature; explanation	Deriving graphs involving water level – time graphs with reasoning			

1. Conversion of Units

Practice 1

Complete the following exercises. Show clear working.

- (a) Convert 36 m/s into km/h.
- (b) Convert 240 km/h into m/s.
- (c) Convert 100 m/s into km/h.
- (d) Convert 56 km/h into m/s.

$$\begin{array}{l|l} 1 \text{ km} = 1000 \text{ m} & 1 \text{ m} = \frac{1}{1000} \text{ km} \\ 1 \text{ m} = 100 \text{ cm} & 1 \text{ cm} = \frac{1}{100} \text{ m} \end{array}$$

Memorise

$$\left. \begin{array}{l} 1 \text{ h} = 60 \text{ min} \\ 1 \text{ min} = 60 \text{ sec} \end{array} \right\} \begin{array}{l} 1 \text{ h} = \frac{3600}{1} \text{ sec} \\ 1 \text{ s} = \frac{1}{3600} \text{ h} \end{array}$$

What prior knowledge do you need to complete the above exercises?

$$(a) \quad 36 \text{ m/s} = \frac{36 \text{ m}}{1 \text{ s}} = \frac{36 \div 1000 \text{ km}}{1 \div 3600 \text{ h}} = \underline{129.6} \text{ km/h}$$

$$(b) \quad \frac{240 \text{ km}}{1 \text{ h}} = \frac{240 \times 1000 \text{ m}}{1 \times 3600 \text{ s}} = 66\frac{2}{3} \text{ m/s}$$

$$(c) \quad \frac{100 \text{ m}}{1 \text{ s}} = \frac{100 \div 1000 \text{ km}}{1 \div 3600 \text{ h}} = 360 \text{ km/h}$$

$$(d) \quad \frac{56 \text{ km}}{1 \text{ h}} = \frac{56 \times 1000 \text{ m}}{1 \times 3600 \text{ s}} = 15\frac{5}{9} \text{ m/s}.$$



Big idea



A graph is a **DIAGRAM** showing the relation between variable quantities, typically of two variables, each measured along one of two axes at right angles.

The graphs we will be looking at for this lesson will be made up of two number lines. A horizontal number line forms the x -axis and a vertical number line forms the y -axis.

What do we mean by the gradient of a graph?

steepness of slope.

How do we calculate the gradient?

$$\text{gradient} = \frac{\text{rise}}{\text{run}}$$

What is the unit of the gradient if unit of x -axis is second(s) and unit of y -axis is metre (m)?

$$\text{Gradient} = \frac{\text{m}}{\text{s}} = \text{m/s}$$

↓
speed

In academic language: The gradient of a function measures the sensitivity to change of the value with respect to the argument.

2. What is Kinematics?

The beginning of the word, 'kinemat-' is Greek and means 'motion.' The end of the word, '-ics,' is Latin and means 'the study of.' Putting it together, **kinematics** is the study of motion.

Kinematics is the science of describing the *motion* of objects using words, diagrams, numbers, graphs, and equations, without considering the causes for the motion.

Why use graphs?

Describing the motion of an object is occasionally hard to do with words. Sometimes graphs help make motion easier to picture and therefore understand.

Distance: total length covered by a moving object, regardless of the direction of motion

Speed: distance travelled per unit of time

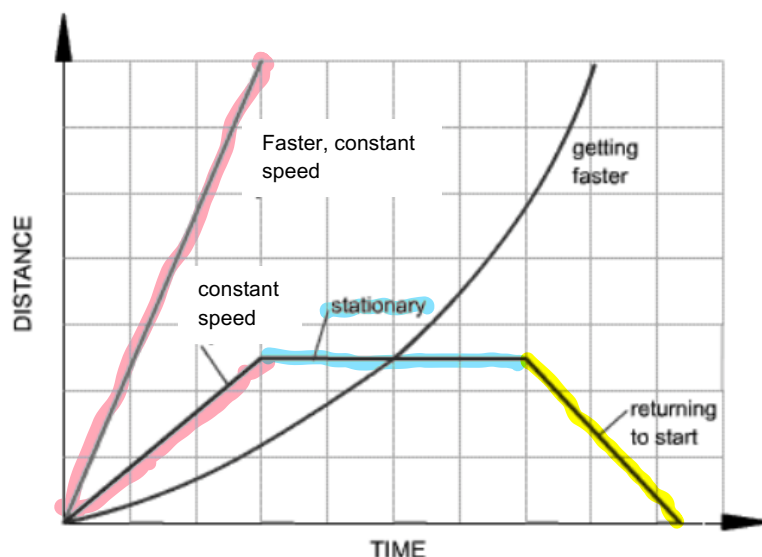
3. Distance-Time Graph

The shape of a Distance-time graph

When an object is in motion, its distance travelled and the time taken to cover that distance can be demonstrated in a **distance-time graph**.

A distance-time graph tells us how far an object has moved with time.

- The steeper the graph, the faster the motion.
- A horizontal line means the object is not changing its position - it is not moving, it is at rest. / stationary $\rightarrow$ speed = 0
- A downward-sloping line means the object is returning to the start.



(Graph from: <http://www.bbc.co.uk/schools/gcsebitesize/physics/forces/speedvelocityaccelerationfhrev2.shtml>)

Discussion

A runner completes a 400 meter lap in 50 seconds. He moves at different speeds at various points during the 400 meter lap. Is his average speed the same as his instantaneous speed (speed at a particular instant)? Yes / **No**

What is the difference between **INSTANTANEOUS SPEED** and **AVERAGE SPEED**?

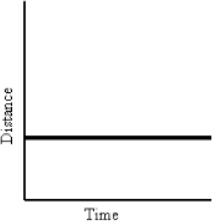
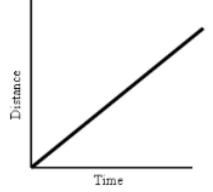
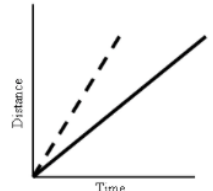
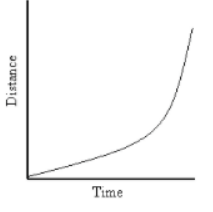
↓
speed at a certain time

↓
 $\frac{\text{Total dist}}{\text{Total time}}$

ACTIVITY 2

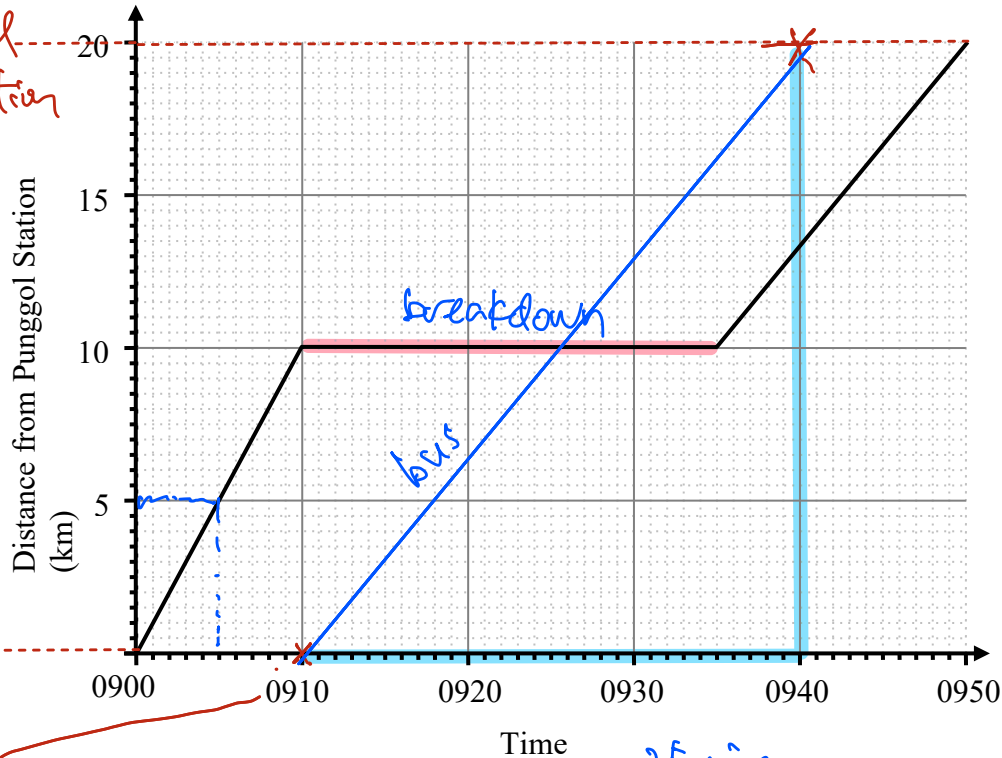
You can access the activity at this link: <https://www.geogebra.org/calculator/jn4zx6tg>

Discussion: How do the graphs differ? Explain.

	Time is increasing to the right, but its distance does not change. It is not moving. We say it is <u>stationary / at rest.</u> Speed is <u>0</u>. Gradient is <u>0</u>.
	Time is increasing to the right, and distance is increasing constantly with time. The object moves at a <u>constant speed.</u> Gradient is <u>constant</u>.
	A steeper line indicates a larger distance moved in a given time. In other words, <u>faster / greater</u> speed. Both lines are straight, so both speeds are <u>constant.</u>
	Graphs that show <u>acceleration</u> look different from those that show constant speed. The line on this graph is curving upwards. This shows an <u>increase</u> in speed since the line is getting steeper. Gradient is <u>increasing</u>. In other words, the object is <u>moving faster.</u>
What can you conclude about the speed and the gradient of the distance-time graph? <u>Speed = gradient of the distance-time graph.</u>	

Practice 2

The graph shows the journey of a train from Punggol Station to Orchard Station.



- How long did the train service break down? *25 mins.*
- How far is the train from Orchard Station after 5 minutes? *$20 - 5 = 15 \text{ km}$*
- How much time would the train need to reach Orchard Station if it has not broken down? *$50 \text{ min} - 25 \text{ min} = 25 \text{ min}$*
- A bus is also travelling from Punggol Station to Orchard Station. It left Punggol station at 0910 and travelled at a constant speed of 40 km/h. Show the journey of the bus on the above graph.

draw a line

- need 2 points.

mark the point

gradient = speed

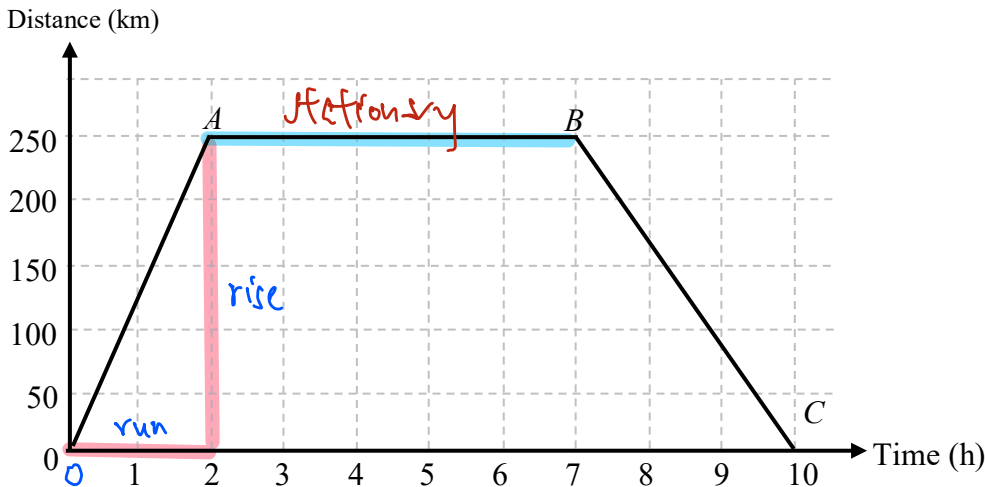
$$= \frac{40 \text{ km}}{1 \text{ h}} \quad \text{rise} \quad \text{run}$$

$$= \frac{20 \text{ km}}{\frac{1}{2} \text{ h}}$$

scale down to mark the point

Practice 3

The distance-time graph shows a journey made by a driver. He started his journey from home at 0730, drove to a town 2 hours later and returned home.



- How long was his car stationary? *5h*
- At what time did he reach home? *1730*
- Calculate his average speed for the whole journey.
- Find the gradient of line segment OA, stating clearly what this gradient represents.

(c) Ave. speed = $\frac{\text{Total dist}}{\text{Total time}}$
 $= \frac{500}{10}$
 $= 50 \text{ km/h}$

(d) Grad = $\frac{\text{rise}}{\text{run}}$
 $= \frac{250}{2}$
 $= 125$

It represents the constant speed of the car travelling at *125 km/h*.

4. Speed-Time Graph



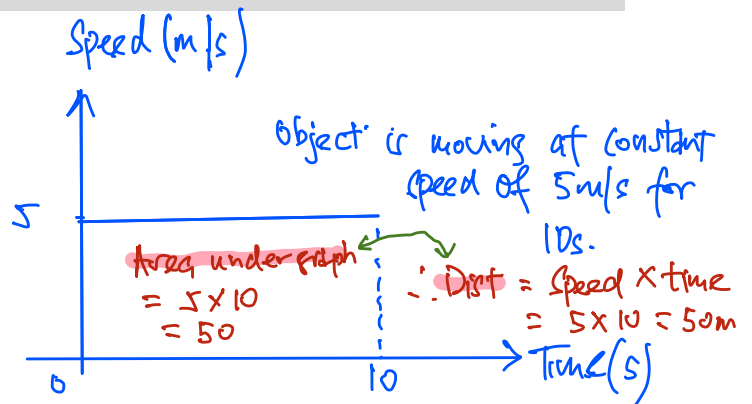
Distance = Speed $\times$ Time



Time = $\frac{\text{Distance}}{\text{Speed}}$



Speed = $\frac{\text{Distance}}{\text{Time}}$



Area under the speed-time graph gives the distance travelled.

Acceleration: Rate of change of speed or gradient of a speed-time graph

An object is experiencing deceleration (retardation) if the gradient of a speed-time graph is negative.

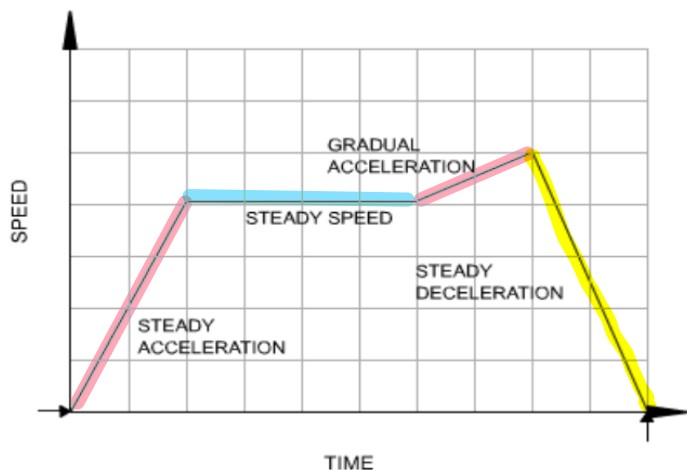
line going down

Acceleration — line going up.

MEMORISE!

A speed-time graph shows us how the speed of a moving object changes with time.

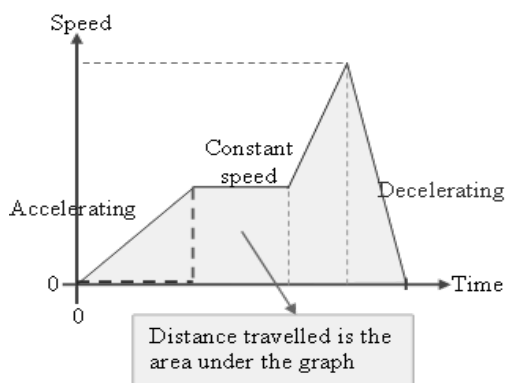
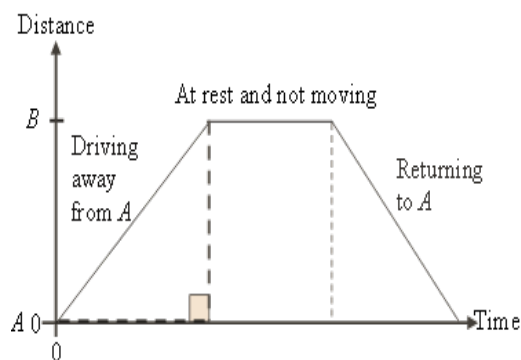
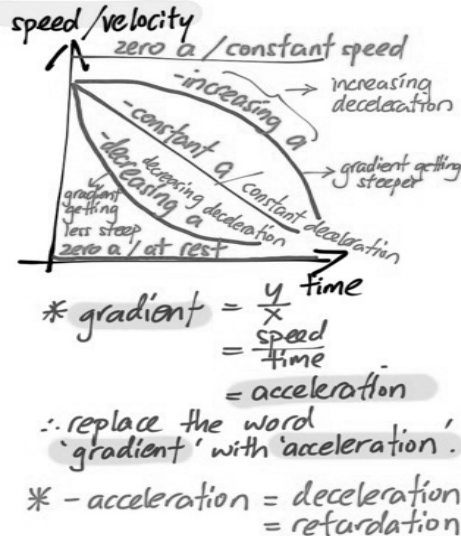
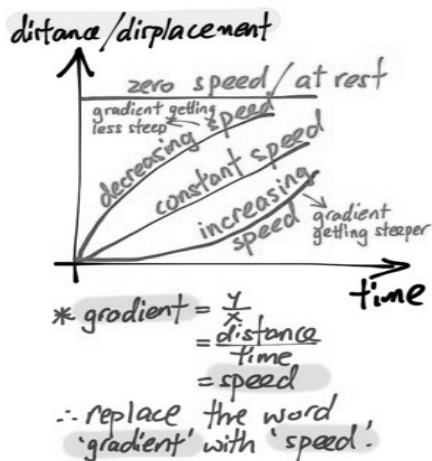
- The **steeper the graph**, the **greater the acceleration**.
- A **horizontal line** means the **object is moving at a constant speed**.
- A downward-sloping line means the object is slowing down.



$$\text{Average Speed} = \frac{\text{Total Distance Travelled}}{\text{Total Time Taken}}$$

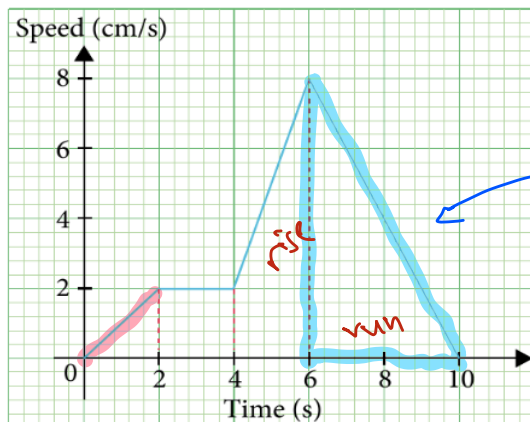
Interpreting Motion of Object from respective Graphs

Using graphical illustrations to summarise the key points covered in this chapter.



Worked Example 1

The graph shows the speed of an object over a period of 10 seconds.



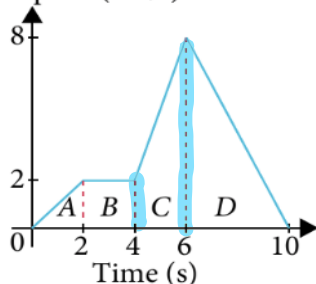
- Find the **acceleration** of the object in the first 2 seconds.
- Given that the distance travelled is given by the area under the speed-time graph, find the average speed of the object during the whole journey.
- Find the **deceleration** of the object in the last 2 seconds.

Solution

$$\begin{aligned} \text{(i) Acceleration} &= \frac{2 \text{ cm/s}}{2 \text{ s}} \\ &= 1 \text{ cm/s}^2 \end{aligned}$$

$$\begin{aligned} \text{Average speed} &= \frac{\text{total distance}}{\text{total time}} \\ &= \frac{32}{10} \\ &= 3.2 \text{ cm/s} \end{aligned}$$

(ii) Speed (cm/s)



$$\begin{aligned} \text{(iii) Deceleration in the last 2 seconds} \\ &= \text{deceleration in the last 4 seconds} \\ &= \frac{8-0}{10-6} \\ &= 2 \text{ cm/s}^2 \end{aligned}$$

Total distance

= area under graph

= area of (A + B + C + D)

$$= \left(\frac{1}{2} \times 2 \times 2 \right) + (2 \times 2) + \frac{1}{2} (2+8) \times 2 + \left(\frac{1}{2} \times 8 \times 4 \right)$$

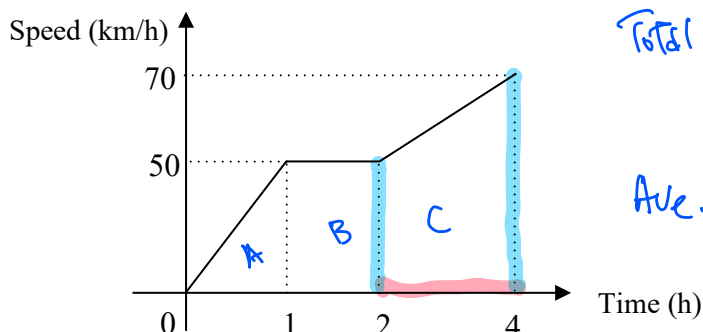
$$= 2 + 4 + 10 + 16$$

$$= 32 \text{ cm}$$

Area of trapezium = $\frac{1}{2} (\text{sum of // sides}) h$

Practice 4

The diagram shows the speed-time graph of a car. Find the average speed for the whole journey.



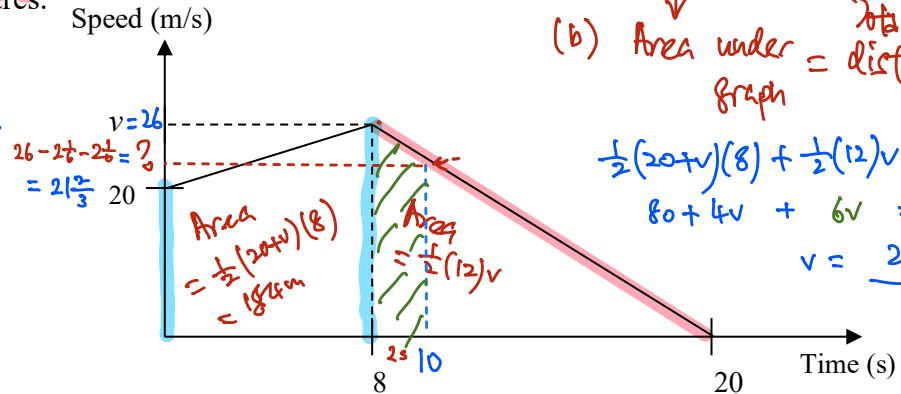
$$\begin{aligned} \text{Total dist} &= \left(\frac{1}{2} \times 1 \times 50 \right) + (1 \times 50) + \left(\frac{1}{2} \times (50+70) \times 2 \right) \\ &= 195 \text{ km.} \end{aligned}$$

$$\begin{aligned} \text{Ave. speed} &= \frac{\text{Total dist}}{\text{Total time}} \\ &= \frac{195}{4} \\ &= 48.75 \text{ km/h.} \end{aligned}$$

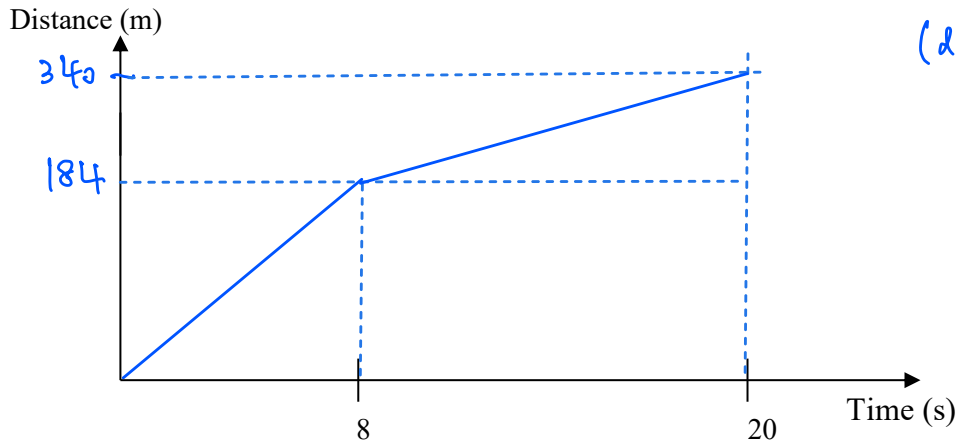
Practice 5

The speed-time graph of a particle is given in the following diagram. The total distance travelled over 20 seconds is 340 metres.

(a) Avg. speed = $\frac{\text{Total dist}}{\text{Total time}}$
 $= \frac{340}{20}$
 $= 17 \text{ m/s}$

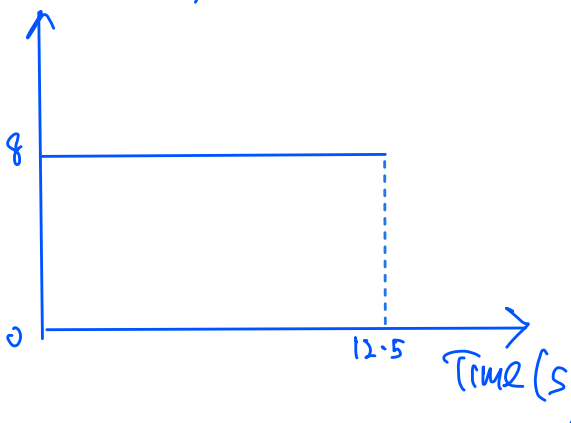
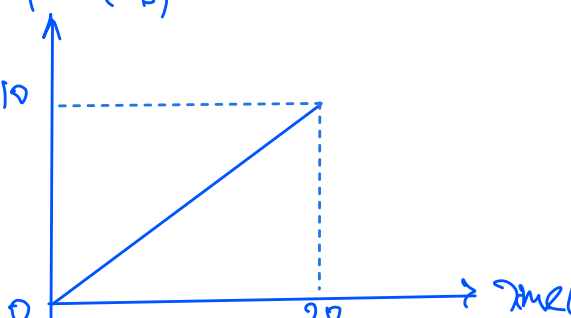


- (a) Find the average speed over 20 seconds.
 (b) Find the greatest speed, v .
 (c) Find the deceleration in the last 12 seconds. Gradient = Deceleration = $\frac{26}{12} = 2\frac{1}{6} \text{ m/s}^2$
 (d) Find the distance travelled in the first 10 seconds.
 (e) On the axes provided below, sketch the distance-time graph for the same journey. ↓ decrease in speed (= $2\frac{1}{6} \text{ m/s}$ per second.)



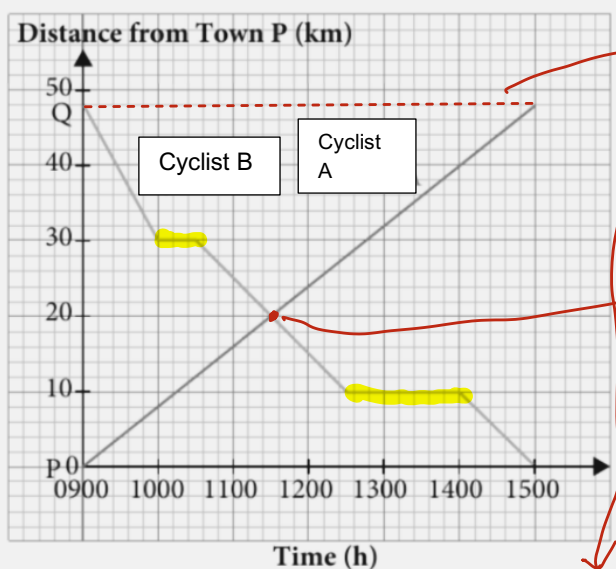
(d) Distance travelled from $t = 8$ to 10 s
 $= \frac{1}{2}(26 + 20/3)(2)$
 $= 47\frac{2}{3}$
 Total distance for first 10s
 $= \frac{1}{2}(20 + 26)(8) + 47\frac{2}{3}$
 $= 184 + 47\frac{2}{3}$
 $= 231\frac{2}{3} \text{ m}$

Practice 6

Bob ran from point <i>A</i> to point <i>B</i> with a constant speed of 8 m/s in 12.5 s. <i>horizontal line</i>	Bob ran from point <i>P</i> to point <i>R</i> in 20 s. His speed at point <i>P</i> is 0 m/s. His speed at point <i>R</i> is 10 m/s. <i>line with positive gradient</i>
(a) Sketch a speed-time graph representing this motion. Speed (m/s)  Time (s)	(a) Given that his acceleration is constant, sketch a speed-time graph representing this motion. Speed (m/s)  Time (s)
(b) What is his acceleration? (Unit: m/s²) 0 m/s^2	(b) What is his acceleration? (Unit: m/s²) <i>gradient</i> $= \frac{10}{20}$ $= \frac{1}{2} \text{ m/s}^2$
(c) Calculate the distance he ran. <i>Area under graph</i> $= 8 \times 12.5 = 100 \text{ m}$	(c) Calculate the distance he ran. <i>Area under graph</i> $= \frac{1}{2} \times 20 \times 10$ $= 100 \text{ m}$

Practice 7

The distance-time graphs show the journeys of Cyclist A and Cyclist B. Cyclist A travelled from Town P to Town Q while Cyclist B travelled from Town Q to Town P.



Don't just look at values on the axis.
Look at the graphs.

They met.

From the graph, find

- the distance between Town P and Town Q,
- the duration Cyclist B took to rest during the journey,
- the distance of both cyclists from Town Q when they met,
- the period of time for which Cyclist B travelled the fastest, steepest gradient.
- the average speed of Cyclist B for the whole journey.

48 km

2 h

$48 - 20 = 28$ km

0900 - 1000

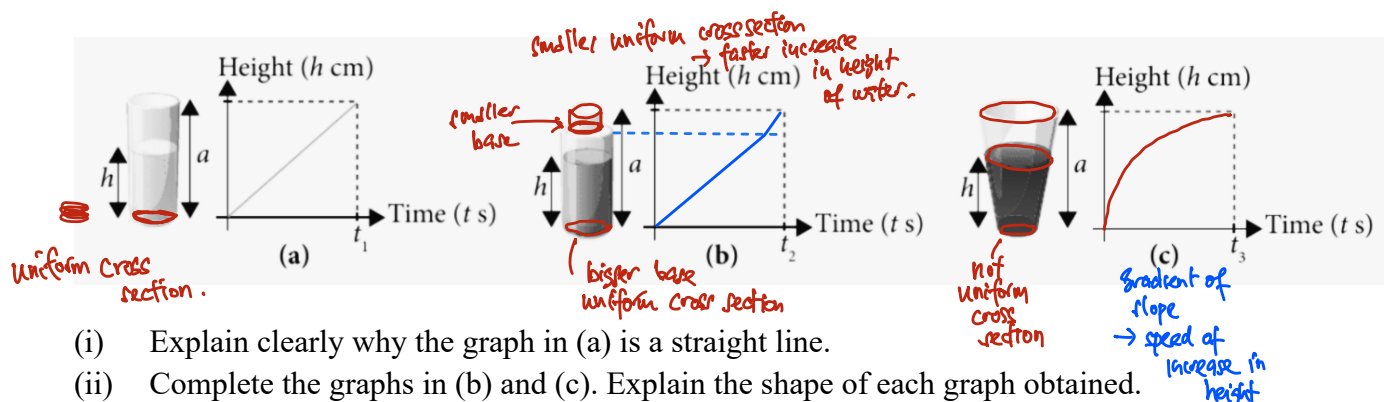
$$\frac{\text{Total dist}}{\text{Total time}} = \frac{48}{6}$$

$$= 8 \text{ km/h.}$$

5. Water Level-Time Graph

Worked Example 2

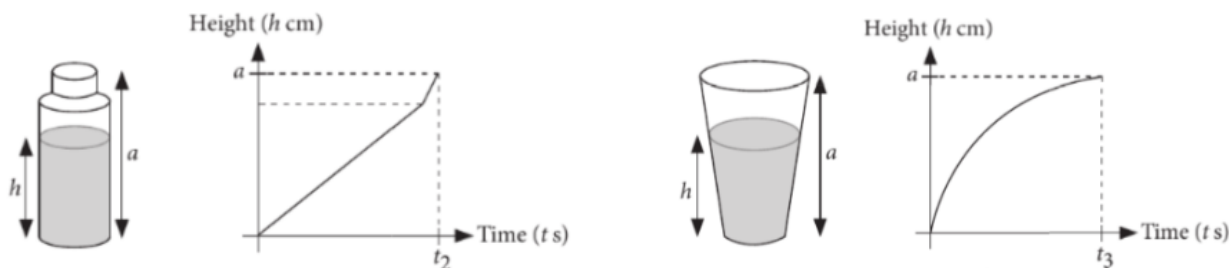
Three containers, that were initially empty, are each being filled with liquid to a height of a cm, by a tap at a constant rate. The graph in (a) shows the height (h cm) of the liquid as the container is being filled in t seconds.



- Explain clearly why the graph in (a) is a straight line.
- Complete the graphs in (b) and (c). Explain the shape of each graph obtained.

Solutions

- The graph is a straight line because the container has a uniform cross-section.
-

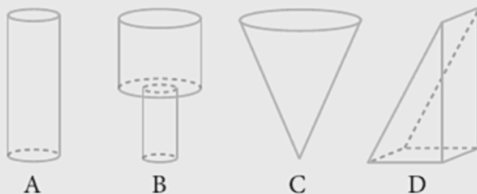


Graph (b) consists of two straight lines, of which the second line is steeper because the container can be segmented into two parts of different uniform cross-sectional area and the bottom part of the container has a larger cross-sectional area than the top part.

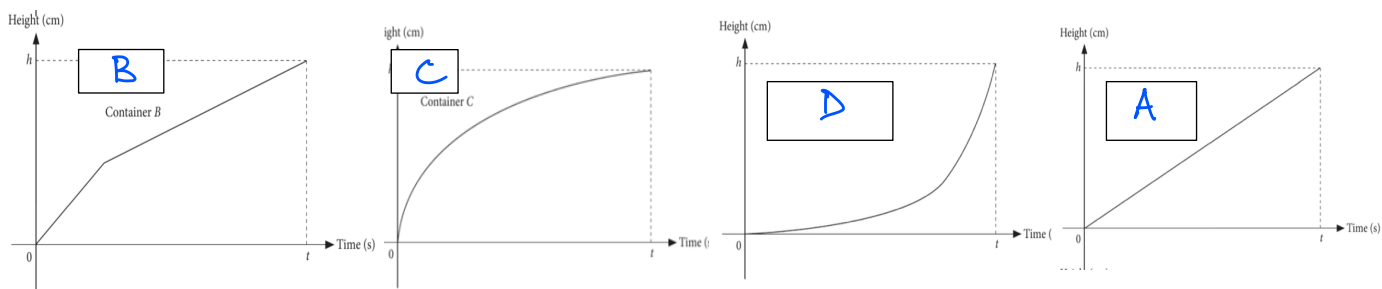
Since the cross-sectional area of the container is non-uniform and increases as h increases, graph (c) consists of a curve that becomes less steep as the container gets wider at the top.

Practice 8

The diagrams below show four containers (not drawn to scale), i.e. A, B, C and D, each with a height of h cm. The containers are initially empty. It takes t seconds to fill each container with water at a constant rate.

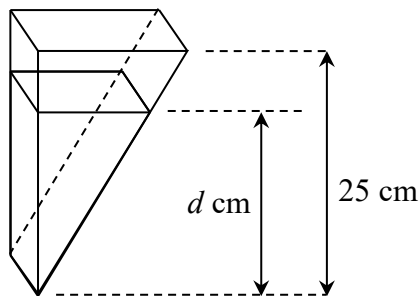


Match the graphs of the depth of the water against time to the containers.



Practice 9 (Challenging)

A container is a prism with a triangular cross-section. The container has a height of 25 cm. Water is poured into the empty container at a constant rate. It takes 10 seconds to fill the container. After t seconds the depth of the water is d cm.



$$\left(\frac{d}{25}\right)^2 = \frac{t}{10} \quad \text{— need next chapter.}$$

- Explain why $\left(\frac{d}{25}\right)^2 = \frac{t}{10}$.
- Find the value of t when $d = 10$.
- Sketch the graph showing how the depth varies during the 10 seconds.

Distance - Time Graph

- Gradient $\rightarrow$ Speed.
- Horizontal line $\rightarrow$ at rest / stationary.

Speed - Time Graph

- Gradient $\rightarrow$ Acceleration / Deceleration
- Horizontal line $\rightarrow$ constant speed
- Distance $\rightarrow$ Area under graph.